

Combinatorics – 8

Permutations, binary recursions and log-concavity

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Theoretical minimum. A statistic on permutations is studied through its generating polynomial, a function given by $f(2n)$ and $f(2n + 1)$ by induction along the binary expansion, and an inequality between neighbouring terms through real roots or a product. Sign and cycle type are in Abstract algebra – 3, generating functions in Combinatorics – 2, Stirling numbers in Combinatorics – 3.

Definitions. For $\sigma \in S_n$ let $\text{inv } \sigma = \#\{i < j : \sigma(i) > \sigma(j)\}$, $\text{des } \sigma = \#\{i < n : \sigma(i) > \sigma(i + 1)\}$ (descents), $\text{exc } \sigma = \#\{i : \sigma(i) > i\}$ (excedances) and $\text{cyc } \sigma$ the number of cycles. The *Eulerian number* $A(n, k)$ is the number of $\sigma \in S_n$ with k descents, $A_n(x) = \sum_k A(n, k)x^k$. A *complete mapping* of an abelian group G is a bijection $\sigma: G \rightarrow G$ such that $x \mapsto x + \sigma(x)$ is also a bijection. A sequence $a_0, \dots, a_m \geq 0$ is *log-concave* if $a_k^2 \geq a_{k-1}a_{k+1}$ for $0 < k < m$, and *unimodal* if $a_0 \leq \dots \leq a_j \geq \dots \geq a_m$ for some j ; $s_2(n)$ is the sum of the binary digits of n .

Theorem 1 (generating polynomials). $\sum_{\sigma \in S_n} q^{\text{inv } \sigma} = \prod_{k=1}^n (1 + q + \dots + q^{k-1})$ and $\sum_{\sigma \in S_n} x^{\text{cyc } \sigma} = x(x+1) \cdots (x+n-1)$. *Proof of the first:* the Lehmer code $c_i = \#\{j > i : \sigma(j) < \sigma(i)\} \in \{0, \dots, n-i\}$ is a bijection of S_n onto $\prod_i \{0, \dots, n-i\}$ with $\text{inv } \sigma = \sum_i c_i$. The second is Combinatorics – 3, homework 7.

Theorem 2 (Eulerian numbers). Inserting n into a permutation of $[n-1]$ with k descents keeps k in $k+1$ of the n gaps (the end and the descents) and raises it by one in the others, so $A(n, k) = (k+1)A(n-1, k) + (n-k)A(n-1, k-1)$. Consequently $A(n, k) = A(n, n-1-k)$,

$$\sum_{m \geq 0} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}} \quad \text{and} \quad x^n = \sum_k A(n, k) \binom{x+k}{n} \quad (\text{Worpitzky}).$$

Foata's transformation (write each cycle from its largest element, order the cycles by increasing largest elements, erase the parentheses) turns $\#\{i :$

$\sigma(i) \geq i\}$ into the number of ascents plus one; hence exc has the same distribution as des .

Theorem 3 (complete mappings). If a finite abelian group G has a complete mapping, then $\sum_{g \in G} g = 0$, because $\sum_x (x + \sigma(x)) = 2 \sum_g g$ must equal $\sum_g g$. The sum is non-zero exactly when G has a unique element of order 2; so $\mathbb{Z}_n$ has a complete mapping if and only if n is odd ($\sigma = \text{id}$ works then): for even n no permutation σ of $\mathbb{Z}_n$ makes $\sigma(i) + i$ a permutation. (Hall–Paige: a finite group has one if and only if its Sylow 2-subgroups are trivial or non-cyclic.)

Theorem 4 (log-concavity). (a) If $\sum_{k \leq m} a_k x^k$ with $a_k \geq 0$ has only real roots, then $a_k^2 \geq a_{k-1} a_{k+1} \left(1 + \frac{1}{k}\right) \left(1 + \frac{1}{m-k}\right)$ (Newton; Polynomials – 4). (b) A log-concave sequence of positive numbers is unimodal, and the convolution of two such sequences is again log-concave: they are the sequences whose Toeplitz matrix (a_{j-i}) has all 2×2 minors ≥ 0 , a property preserved by products (Cauchy–Binet). Thus $\binom{n}{k}$, $c(n, k)$ and $A(n, k)$ are log-concave in k , the last because A_n has only real roots (homework 1).

Fact 1 (binary recursions). Statements about f given through $f(2n)$, $f(2n+1)$ are proved by induction split by parity, or over blocks $[2^m, 2^{m+1})$. Thus $s_2(2n+1) = s_2(n) + 1$ gives $v_2(n!) = n - s_2(n)$ and $\sum_{k < 2^m} s_2(k) = m2^{m-1}$; $\sum_{k < 2^m} (-1)^{s_2(k)} P(k) = 0$ whenever $\deg P < m$ (Prouhet, Thue–Morse); and Stern’s sequence $s(0) = 0$, $s(1) = 1$, $s(2n) = s(n)$, $s(2n+1) = s(n) + s(n+1)$ has coprime neighbours, $s(n+1)$ being the number of ways to write n as a sum of powers of 2 used at most twice.

Fact 2 (parity via involutions). If ι is an involution of a finite set X , then $|X| \equiv |\text{Fix } \iota| \pmod{2}$. Useful involutions: reversing a word, swapping two coordinates, $\sigma \mapsto \sigma^{-1}$; a count $\text{tr } M^N$ of closed walks may be computed with M reduced modulo 2.

1. Prove the first formula of Theorem 1 via the Lehmer code and deduce that the number of permutations of $[n]$ with k inversions is symmetric and unimodal in k . Prove the recurrence of Theorem 2, derive from it $\sum_{m \geq 0} (m+1)^n t^m = A_n(t)/(1-t)^{n+1}$, and obtain Worpitzky’s identity by comparing coefficients. Compute $A_n(x)$ for $n \leq 5$.

2. Prove Theorem 3 for finite abelian groups. Show that $x \mapsto ax$ is a complete mapping of $\mathbb{Z}_n$ if and only if $\gcd(a(a+1), n) = 1$, find a complete mapping of $\mathbb{Z}_2 \times \mathbb{Z}_2$, and show that $\mathbb{Z}_n$ has a complete mapping exactly when there is a Latin square of order n orthogonal to its addition table.
3. For a positive integer n , let $f(n)$ be the number obtained by writing n in binary and replacing every 0 with 1 and vice versa. For example, $n = 23$ is 10111 in binary, so $f(n)$ is 1000 in binary, therefore $f(23) = 8$. Prove that

$$\sum_{k=1}^n f(k) \leq \frac{n^2}{4}.$$

When does equality hold?

(IMC 2015)

4. Let k be a positive integer. For each non-negative integer n , let $f(n)$ be the number of solutions $(x_1, \dots, x_k) \in \mathbb{Z}^k$ of the inequality $|x_1| + \dots + |x_k| \leq n$. Prove that for every $n \geq 1$ we have $f(n-1)f(n+1) \leq f(n)^2$. (The symmetry of f in n and k is Combinatorics – 2, homework 8.) (IMC 2016)
5. Fix positive integers n and k such that $2 \leq k \leq n$ and a set M consisting of n fruits. A permutation is a sequence $x = (x_1, x_2, \dots, x_n)$ such that $\{x_1, \dots, x_n\} = M$. Ivan prefers some (at least one) of these permutations. He realized that for every preferred permutation x , there exist k indices $i_1 < i_2 < \dots < i_k$ with the following property: for every $1 \leq j < k$, if he swaps x_{i_j} and $x_{i_{j+1}}$, he obtains another preferred permutation. Prove that he prefers at least $k!$ permutations. (IMC 2023)